Logs, Traces, Metrics, Oh My!

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Setting Expectations

What this is:

- A short overview of some basic software observability principals
- An introduction to how OpenTelemetry (OTel) maps to those principals

What this isn't:

- A deep dive into OpeneTelemetry Protocol (OTLP)
- An in depth talk about observability in production

Overview

Topics we will cover:

- What is Software Observability?
  - Why does it matter?
- What is OpenTelemety?
  - How does it differ from other forms of Observability?
- What is the OTel Collector?
- A short explanation of the system diagram and what each piece is.

What is Software Observability?

Observability is being able to inspect the runtime state of our software. It allows us to see what our software is doing and when. The data that describes this is called telemetry.

The 3 most common types of telemetry used in modern software are:

- Logs
- Metrics
- Traces

Metrics

Metrics are heuristics about our software that is measured at single points in time. This data can then be aggregated and visualized over time. Some examples include:

- CPU Usage
- Total HTTP Requests
- Error Rate

Traces

Traces are information about how software is invoked and what the flow of requests and data looks like. The base unit of a trace is the span. They can be heirarchical and create trees of related spans that you can attach metadata to. Examples include:

- Tracking what services get invoked when you make a particular HTTP request
- Finding where an error occurred in the chain of calls from a request
- Checking latency from remote services such as databases or 3rd party API's

Logs

Logs are the most common form of telemetry. This is usually where people start due to how easy they are to generate and capture. However, logs are generally best used as a suppliment to metrics and traces. For example:

- Look at the surrounding logs from where a trace errored and see if there was anything out of the ordinary
- Correlating logs with specific HTTP status codes to see what the system was doing when it errored

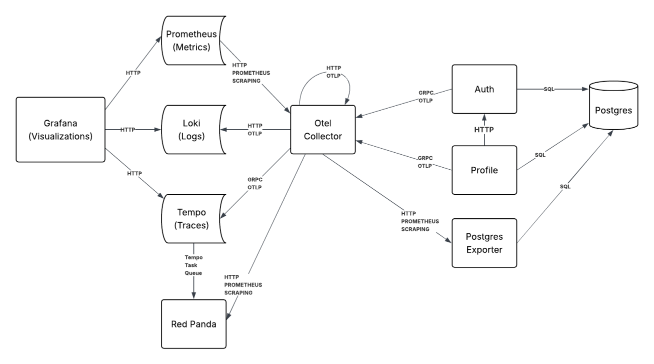
What is OpenTelemetry (OTel)?

> OpenTelemetry is an open source observability framework for cloud native software. It provides a single set of APIs, libraries, agents, and collector services to capture distributed traces and metrics from your application.

Put simply, OTel is a set of standards, tools, and SDK's for handling telemetry data. There are 3 main components that OTel manage:

- OTLP
  - The OpenTelemetry Protocol. This is a specification for how OTel data should be shaped and transported.
- OTel SDK's
  - Reusable libraries that you can use to implement OTLP based observability.
- OTel Collector
  - Vendor-agnostic way to receive, process and export telemetry data.

System Diagram



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